

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Information Technology) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BITIPE603E - DATA WAREHOUSING & DATA MINING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which schema uses a central fact table connected to multiple dimension tables?
A. Snowflake schema
B. Galaxy schema
C. Hierarchical schema
D. Star schema
 - 2 What does “granularity” in a data warehouse refer to?
A. Number of columns in a table
B. Level of detail of the stored data
C. Number of users accessing the system
D. Data accuracy level
 - 3 A key difference between data mining and OLAP is:
A. OLAP predicts trends; data mining doesn't
B. Data mining finds hidden patterns; OLAP summarizes known data
C. OLAP is for real-time applications; data mining is not
D. Data mining uses more memory
 - 4 During ETL (Extract, Transform, Load), surrogate keys are introduced in the dimension tables. Why is this critical for data mining applications?
A. Surrogate keys increase primary key integrity
B. They prevent duplication of transactional data
C. They provide consistency across historical versions and SCDs
D. They improve user-level interpretability of data
 - 5 In OLAP, which operation is used to increase the granularity of data?
A. Roll-up
B. Slice
C. Dice
D. Drill-down

- 6 Which of the following techniques is typically applied in data reduction to maintain the integrity of patterns in large datasets?
 - A. Data normalization
 - B. Principal Component Analysis (PCA)
 - C. Noise filtering
 - D. Concept hierarchy generation
- 7 What distinguishes classification from prediction in data mining?
 - A. Classification always uses regression
 - B. Prediction deals with categorical outputs
 - C. Classification maps data to predefined labels; prediction estimates continuous values
 - D. Prediction requires supervised learning only
- 8 Which of the following is an example of an association rule?
 - A. IF income = high THEN buy = yes
 - B. 10% of customers bought product A
 - C. Mean purchase = \$50
 - D. Customers are classified as loyal
- 9 A hospital uses patient history to predict the likelihood of heart disease. This is an example of:
 - A. Clustering
 - B. Classification
 - C. Association
 - D. Outlier Detection
- 10 In smart grids, data mining is used to forecast electricity demand based on historical usage and weather data. This is a case of:
 - A. OLAP slicing
 - B. Regression
 - C. Association rule mining
 - D. Classification
- 11 What challenge does spatial data mining face compared to traditional data mining?
 - A. Limited scalability
 - B. Complex data types like polygons and trajectories
 - C. Text format compatibility
 - D. Fixed schema requirements
- 12 Which of the following is a major social concern regarding data mining?
 - A. High power consumption
 - B. Violation of user privacy
 - C. Network latency
 - D. Duplicate backups

Q2. Solve the following.

- A) Explain Data mining as a step in KDD. 6
- B) Explain different types of attributes with suitable example. 6

Q3. Solve the following.

- A) Use these methods to normalize the following group of data: 6
 200, 300, 400, 600, 1000
 (a) min-max normalization by setting min = 0 and max = 1
 (b) z-score normalization
 (c) normalization by decimal scaling

- B) Find the outliers, if any, for the following data set, and draw the box-and-whisker plot. Mark any outliers with an asterisk. 6

21, 23, 24, 25, 29, 33, 49

Q4. Solve Any Two of the following.

- A) With the help of algorithm, explain decision tree induction for classification. 6
 B) Transactions consist of a set of items $I = \{a, b, c, \dots\}$, min-support=3. Draw FP Tree. 6

TID	Items Bought
1	f, a, c, d, g, i, m, p
2	a, b, c, f, i, m, o
3	b, f, h, j, o
4	b, c, k, s, p
5	a, f, c, e, i, p, m, n

- C) Consider the five transactions given below. If minimum support is 30% and minimum confidence is 80%, determine the frequent itemsets and association rules using the apriori algorithm. 6

Transaction	Items
T1	Bread, Jelly, Butter
T2	Bread, Butter
T3	Bread, Milk, Butter
T4	Coke, Bread
T5	Coke, Milk

Q5. Solve Any Two of the following.

- A) Explain Agglomerative and Divisive Hierarchical Clustering Methods. 6
 B) Use K-means algorithm to create 3-clusters for given set of values: 6
 $\{2, 3, 6, 8, 9, 12, 15, 18, 22\}$
 C) Predict a class label of an unknown sample $X = \{\text{age} = \leq 30, \text{Income} = \text{"Medium"}, \text{Student} = \text{"Yes"}, \text{Credit_rating} = \text{"Fair"}\}$, using Naïve Bayesian classification on the following training dataset from all electronics customer database. 6

Age	Income	Student	Credit_rating	Class: buys_computer
≤ 30	High	No	Fair	No
≤ 30	High	No	Excellent	No
31..40	High	No	Fair	Yes
>40	Medium	No	Fair	Yes
>40	Low	Yes	Fair	Yes
>40	Low	Yes	Excellent	No
31..40	Low	Yes	Excellent	Yes
≤ 30	Medium	No	Fair	No
≤ 30	Low	Yes	Fair	Yes
>40	Medium	Yes	Fair	Yes
≤ 30	Medium	Yes	Excellent	Yes
31..40	Medium	No	Excellent	Yes
31..40	High	Yes	Fair	Yes
>40	Medium	No	Excellent	No

Q6. Solve Any Two of the following.

- A) Write a note on : Mining Text Databases 6

- B) Differentiate between single linkage, average linkage and complete linkage algorithms. 6
Consider the data given below. Create adjacency matrix.

Object	Attribute 1(X)	Attribute 2(Y)
A	2	2
B	3	2
C	1	1
D	3	1
E	1.5	0.5

- C) What is Web Structure Mining? List the approaches used to structure the web pages to improve on the effectiveness of search engines and crawlers. Explain Page Rank technique in detail. 6

*** End ***